

NESEN AZ YALCIN

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EDUCATION

Sabancı University

B.S. in Computer Science and Engineering

GPA: 3.65/4.0

Relevant Coursework: Software Engineering, Data Structures, Data Science, Artificial Intelligence, Natural Language Processing, Mobile Application Development

Oct 2022 – Jun 2026

Istanbul, Turkey

Sabancı University

B.S. in Industrial Engineering

GPA: 3.65/4.0

Relevant Coursework: Operations Research I, II & III, Simulation, Statistical Modelling, Decision Economics, Decision Analysis, Digital Transformation

Oct 2022 – Jan 2027

Istanbul, Turkey

Sungkyunkwan University

Global Exchange Program

Aug 2026 – Dec 2026

Seoul, South Korea

Alma Mater Studiorum Università di Bologna

Erasmus+ Exchange Program

Feb 2025 – Jun 2025

Bologna, Italy

PUBLICATIONS

- **N. Yalçın**, E. Teper, M. Keskin, M. Öztürk Umur, Y. H. Şahin. “Beyond Skewness: Confidence-Guided Rank Fusion for Text Retrieval”. International Conference on Machine Learning and Applications (ICMLA 2026), IEEE (Accepted). [Paper] [Acceptance]
- D. Alkanalka, M. Seçen, **N. Yalçın**, H. E. Yıldız, E. Çimenoglu, T. Helvacioğlu, D. Keküllüoğlu. “Edge AI Integration Architecture for the Internet of Things”. 34th Signal Processing and Communications Applications Conference (SIU 2026), IEEE. [Paper]

RESEARCH EXPERIENCE

University of Twente

Research Intern (Erasmus+ Traineeship)

Jun 2026 – Aug 2026

Enschede, Netherlands

- Reproduced three published AI/ML papers on 5G/O-RAN network security and handover prediction from their written methodology alone, then diagnosed discrepancies against the authors’ released code and models.
- Synthesized results into a nine-point Minimum Disclosure Set for ML paper reproducibility, published to the team’s documentation site.

Siemens

Senior Project Student

Sep 2025 – Jun 2026

Istanbul, Turkey

- Built a resource-efficient, secure Action Orchestration Agent for industrial edge automation using small quantized language models (SLMs) to map natural language operator commands to protocol-driven MQTT actions locally.

Sabancı University – SUHOPE

Undergraduate Research Assistant

Feb 2025 – Jun 2026

Istanbul, Turkey

- Contributed to ongoing research projects at the Sustainable Health and Humanitarian Operations Research Lab (SUHOPE), collaborating with PhD students and faculty on humanitarian logistics and disaster response.

Sabancı University – PURE (Program for Undergraduate Research)

Research Assistant

Feb 2024 – Feb 2025

Istanbul, Turkey

- Conducted research on disaster relief logistics over three semesters under the supervision of Prof. Raha Akhavan Tabatabaei and Altuğ Tanaltay.

INDUSTRY EXPERIENCE

Hepsiburada Data Science Intern	Jan 2026 – Jun 2026 <i>Istanbul, Turkey</i>
- Developed a query labeling framework using LLMs and applied machine learning techniques to classify queries as broad, specific, or ambiguous. - Prototyped agent-based systems to automate experimentation for new search features. - Optimized ranking feature weights using evaluation metrics including NDCG@k and CR@k, directly targeting search-driven sales performance.	
Bekaert R&D Intern	Aug 2025 <i>Shanghai, China</i>
Designed and implemented data extraction and reporting pipelines from .zs2 files for mechanical lab tests, and applied machine learning to classify tensile test outcomes as acceptable or defective.	
Getirfinans Data Science Intern	Jun 2025 – Jul 2025 <i>Istanbul, Turkey</i>
Automated the extraction and comparison of interest and deposit rates with asynchronous Python and Playwright web scraping pipelines.	

TEACHING EXPERIENCE

Sabancı University Learning Assistant, DSA 210 (Introduction to Data Science)	Feb 2026 – Jun 2026 <i>Istanbul, Turkey</i>
- Led weekly office hours and one-on-one meetings with students, providing guidance on course projects from conception to completion. - Supported students through the full project lifecycle, from data collection and analysis to final presentation, gaining firsthand experience in academic mentorship and technical instruction.	

PROJECTS

Resource-Efficient AI Action Orchestration Platform for Industrial Edge Devices	
- Benchmarked small language models (FunctionGemma-270M, Granite-350M, Gemma 3-1B) under zero-shot, few-shot, and fine-tuned settings for translating natural-language operator commands into MQTT tool calls, using a synthetic dataset of 3,700+ commands. - Developed a method-based intent modeling approach that separates action type from device identity, improving scalability across sensor configurations. - Fine-tuned Granite-350M on the curated dataset, raising exact-match test accuracy from 17.0% to 99.5% (100% tool-call accuracy, 96.8% correct rejection of out-of-scope commands) and matching Gemma 3-1B with roughly one-third of the parameters, enabling deployment on 4–8 GB VRAM edge hardware.	
Disaster Relief Logistics: Assessing Multinational Brands’ Responses [Report]	
- Conducted data-driven analysis of earthquake-related tweets using Python for classification, correlation, and regression, revealing higher engagement in disaster-related posts and evaluated brands’ strategic communications. - Identified sector-specific response patterns, from donations and supply deliveries to free communication and logistics support, and evaluated brands’ crisis communication strategies.	
Improving Coordination in Disaster Relief Donations via News Cycle Analysis [Report]	
- Identified a critical coordination gap following the 2023 Kahramanmaraş earthquake: almost all news outlets relied on just two AFAD need lists for eleven affected cities, failing to capture shifting regional needs over time. - Revealed that the top five most communicated needs shifted week by week, from food and water in the first days to shelter and hygiene products by the fourth week, with over-donation of clothing creating mismatches between supply and demand.	

- Compared media-reported needs against donor behavior survey data, finding largely overlapping patterns but notable discrepancies in child-specific and sleeping supply categories.

Corporate Social Responsibility Communication in Disaster Relief [Report]

- Collected and annotated thousands of social media posts from 60 local and international companies to analyze communication strategies during the 2023 Kahramanmaras earthquake.
- Classified content into informational, transformational, and interactional types, revealing that emotionally resonant messaging with visual media achieved the highest public engagement.

Airport Terminal Simulation & Optimization

- Developed a discrete-event simulation model in Arena to analyze passenger flow and optimize resource allocation.
- Validated the model with Chi-Square, Kolmogorov–Smirnov, and paired t-tests.

Real-Time Messaging Application

- Built a messaging and friend management system with Java, Spring Boot, MongoDB, Docker, and JWT authentication, implementing user registration, sessions, and message exchange.

AWARDS & ACHIEVEMENTS

- **Fulbright Principal Candidate (2026–2027):** Selected as a principal nominee by the Turkish Fulbright Commission for the Fulbright Foreign Student Program to pursue graduate studies in the United States.
- **Dean’s High Honor List:** Earned High Honor at Sabanci University for outstanding academic performance for the following semesters: Fall 2022-2023, Fall 2023-2024, Spring 2023-2024, Fall 2024-2025 and Fall 2025-2026.
- **1st Place, Promise for Tomorrow Program:** Recognized for a startup idea in a competitive program organized by Sisterslab, an NGO promoting gender equality in STEM, and Hepsiburada, one of Turkey’s leading e-commerce companies.
- **Earthquake Full Scholarship:** Awarded a full scholarship at Sabanci University for Spring 2022-23, Fall 2023-24, and Spring 2023-24 based on academic merit and the personal impact of the 2023 Kahramanmaras earthquake.

COMMUNITY SERVICE & VOLUNTEERING

School Support Association

- Teaching mathematics to middle school students from underserved regions across Turkey with limited access to educational resources.

Civic Engagement Project

- Participated in a civic engagement initiative supporting refugees, low-income families, and women with limited access to education, raising awareness on social inequality, equal opportunity, and sex education.

Animal Rights Volunteer – Haytap & Local Groups

- Volunteered with the Animal Rights Federation in Turkey (Haytap) and local organizations following the 2023 Kahramanmaras earthquake, supporting injured and displaced animals in affected areas.

SKILLS

- **Programming & Tools:** Python, C++, Java, R, SQL, MongoDB, Git, Docker, MQTT
- **Machine Learning & AI:** Scikit-learn, TensorFlow, PyTorch, NLTK, Transformers (BERT, GPT)
- **Optimization & Simulation:** Arena Simulation Software, Gurobi
- **Languages:** Turkish (Native), English (C2)
- **IELTS:** Overall band score 8.5/9.0 (Listening 9.0, Reading 9.0, Writing 7.5, Speaking 8.0) [Certificate]